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Initially, the angle between the polarising filters is 90° .

After rotating 30° , the angle between them is 60° .

Using Malu's law,

$$I = I_o \cos^2 60^\circ$$

$$\frac{I}{I_o} = \frac{1}{4}$$

Since intensity is proportional to square of amplitude,

$$\frac{A'}{A} = \sqrt{\frac{I}{I_o}} = \frac{1}{2}$$

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